ADDENDUM · SUPERSEDES §03, §04 AND §09 OF THE MAIN REVIEW

Task 0A / 0B Addendum

Five corrections applied before implementation. The replay input is changed from the validated view to the raw register, Task 0 is split in two, and the blocker list is cut from ten items to three.

Correction 01

Replay input source

> CORRECTED — THE ORIGINAL WAS CIRCULAR
>
> The main review listed `gold.energy_consumption_validated` as Task 0's input. That is circular: the view already carries `previous_valid_kwh`, `validated_kwh_consumption`, `consumption_cleaning_status`, `interval_quality` and `gap_minutes` — the exact outputs the replay exists to derive. A replay reading those is not reproducing the §6.7 logic, it is copying its answer.

The replay input is `gold.energy_raw`. The validated view becomes the comparison target only, and is never opened during the replay itself.

ROLE	OBJECT	COLUMNS

Replay input SOURCE ONLY	`gold.energy_raw`	`energy_meter_key`
		`reading_timestamp_utc`
		`active_energy_cumulative_kwh`

Identity join SOURCE ONLY	`gold.meter`	`energy_meter_key → meter_label` needed for the `ZN.13-%` exclusion and the two startup-baseline meter predicates

Comparison target NEVER AN INPUT	`gold.energy_consumption_validated`	`validated_kwh_consumption`
		`interval_quality`
		`gap_minutes`

Four source columns in total. Everything else is **derived in code and never read**: `previous_valid_timestamp`, `previous_valid_kwh`, `validated_kwh_consumption`, `consumption_cleaning_status`, `interval_quality`, `gap_minutes`, and both Saudi timestamp columns (which are a pure UTC+3 conversion, so deriving them also proves the conversion rather than borrowing it).

SUFFICIENCY, VERIFIED ON THE LIVE DATABASE

4,535,918 gold.energy_raw rows	**− 275,069** Zone 13 rows (excluded by rule 1)	**= 4,260,849** exactly the validated row count	**0** NULL cumulative-kWh rows

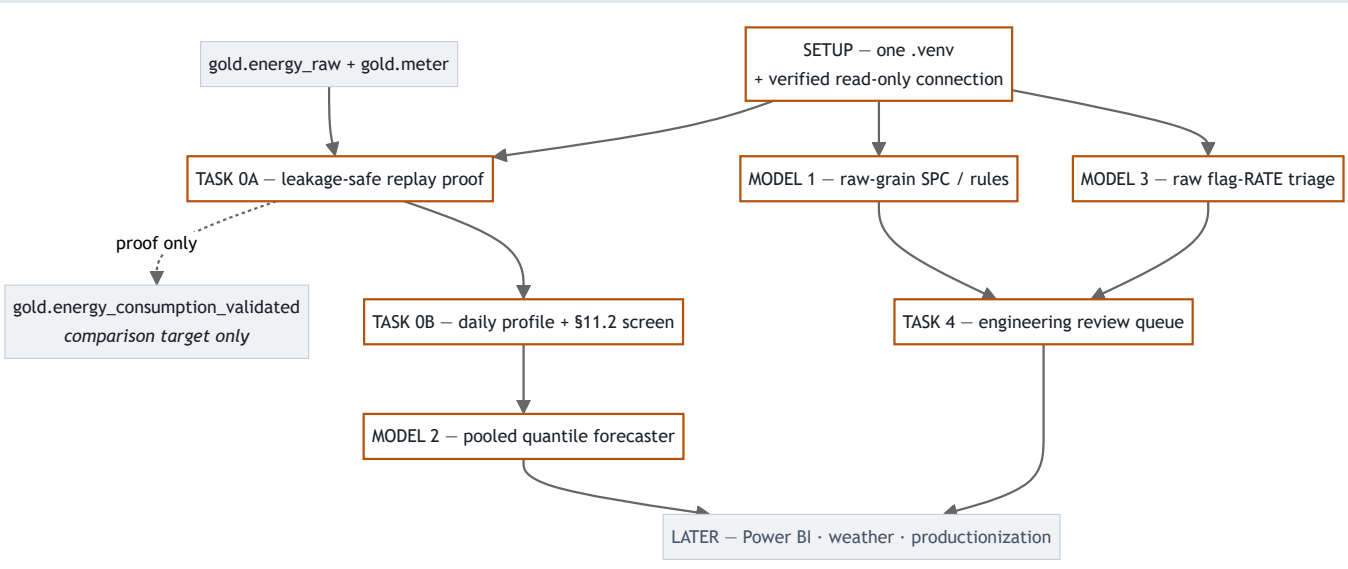
The identity is exact, so the raw register is a complete and sufficient input — and the row count doubles as Task 0A's first reconciliation check.

Correction 02

Task 0 split, and the corrected graph

TASK	PRODUCES	INPUT	BLOCKS
Task 0A Leakage-safe replay	Equivalence proof + `run_metadata.json` divergence figure	`gold.energy_raw` + `gold.meter`	**Nothing.** Proof and forward guard.
Task 0B Daily profile reconstruction	`daily_consumption_profile_v2.csv` + quality screen	Task 0A interval output (cross-checked against the validated view)	**Model 2.**

The distinction the main review collapsed: **0A is not needed for historical forecasting**, because when the forecast is issued at end of Saudi day D the replay reproduces Gold exactly on all 4,260,849 rows. But **0B is needed**, and not because of leakage — because `daily_consumption_profile.csv` and `02_prepare_daily_data.py` are both gone, so the daily training table does not exist and has to be rebuilt before Model 2 has anything to train on.



Model 1 and Model 3 hang off SETUP alone — neither touches Task 0. Model 3 in particular reads raw flag *rates* from `gold.energy_raw` and needs no daily profile, so the `--daily-profile` argument in the run order should be dropped.

Correction 03

Promotion-gate wording

> WITHDRAWN
>
> The main review said the gate is "purely relative to the naive baseline, so a 0.2pp win counts as success." That is wrong. The threshold is **≥ 2.0 percentage points** of WAPE improvement over the previous-day baseline, per meter — a 0.2pp improvement **does not pass**.

The narrower point that does stand: the gate is expressed *only* in relation to the naive baseline, and no document sets an absolute accuracy floor. Pipeline §11.7 records that the charter's ">85% accuracy" target is not mathematically defined and that WAPE must not be relabelled as accuracy. So a meter can clear a 2pp relative margin while still forecasting poorly in absolute terms — the gate answers "better than naive?", not "good enough?".

RETAINED, UNCHANGED

- **Selection happens on validation** — val1 (2026-01-12 → 03-12) and val2 (2026-03-13 → 05-11). The final test window (2026-05-12 → 07-10) stays untouched and is reported once, for whatever the gate already chose.
- **Quantify uncertainty before finalizing the 2pp rule.** At roughly 46 evaluated observations per meter (1,344 common-method observations across 29 meters), the sampling error on a per-meter WAPE is plausibly the same order as the threshold itself. Run a paired bootstrap over meter-days, per meter, and confirm or revise the 2pp figure against that distribution rather than assuming it.

Correction 04

True Task 0 blockers

The main review listed ten decisions and implied six blocked Task 0. Only **three** affect Task 0A/0B code.

#	DECISION	BLOCKS	WHY IT IS A REAL BLOCKER
1	**One Python environment and a verified read-only connection**	0A + 0B	No code runs at all. The venv has psycopg but no lightgbm; base Python has lightgbm but no psycopg; and the `db.py` read-only assert cannot pass as written.

#	DECISION	BLOCKS	WHY IT IS A REAL BLOCKER
2	**Sign-off on the Task 0A input contract** — the four source columns in Correction 01	0A	Determines whether the replay is independent or circular. Proposed above; needs your yes.
3	**Sign-off on the two §11.2 reconstructions** — coverage and boundary days, in Correction 05	0B	They are not recoverable from any surviving artifact and they set the screen that defines Model 2's training set.

MOVED TO THEIR OWNING TASK — NOT TASK 0 BLOCKERS

DECISION	ACTUALLY BLOCKS
Forecast issue time and the lag convention	Model 2 `TASK 2`
Where the promotion gate is evaluated	Model 2 `TASK 2`
Reading `silver.dim_meter_cb_rating` for breaker ratings	Model 2 `TASK 2`
Model 1 acceptance set — 4, 5 or 6 meters	Model 1 `TASK 1`
PF minimum-load gate	Model 1 `TASK 1`
Definition of `severity`	Models 1 and 3, review queue `TASKS 1/3/4`
Model 3's dependency on Task 0	Resolved — there is none `CLOSED`
Power BI promotion	Out of scope `CLOSED`

Correction 05

The two §11.2 reconstructions

(a) Normal-interval duration coverage

Each row in the validated set represents an interval *ending* at `reading_timestamp_utc` and starting at `previous_valid_timestamp`, of length `gap_minutes`. §11.2 states two different rules for the two quantities: consumption is assigned *whole* to the Saudi date of the ending reading, while "duration coverage is split at Saudi-local midnight."

Proposed reconstruction. For meter *m* and Saudi date *D*:

```
clip each interval [previous_valid_timestamp, reading_timestamp_utc]
     to the window [ midnight_saudi(D), midnight_saudi(D+1) )

coverage(m, D) = Σ clipped_seconds  over rows with interval_quality = 'NORMAL_INTERVAL'
                 ─────────────────
                       86,400

screen passes when   0.95 ≤ coverage(m, D) ≤ 1.00
```

Two supports for this reading. Riyadh is UTC+3 with **no DST**, so every day is exactly 86,400 seconds and no special case is needed. And the stated band's upper bound of exactly 100% is only reachable if durations are clipped — an unclipped sum exceeds a full day on any date whose first interval began before midnight, which would make a "100%" ceiling unattainable rather than merely tight.

(b) Per-meter boundary-day identification

Proposed reconstruction. For each meter independently, over *all* rows present in the validated set — including `IGNORED_READING` rows:

```
first_boundary(m) = min(reading_date_saudi)
last_boundary(m)  = max(reading_date_saudi)
```

Verified live: this yields exactly **58** boundary meter-days — 29 meters × 2, with no meter whose first and last dates coincide.

> ONE TRAP, VERIFIED
>
> The boundary must be `min(reading_date_saudi)` over all rows, **not** the date of the meter's `FIRST_VALID_READING`. Those two agree for only **27 of 29** meters: ZN.1-PB.1 and ZN.4-PB.1 carry `STARTUP_EXCLUDED` rows on earlier dates than their first valid reading, so using the first-valid date places their boundary a day late and admits a partial day into the training set.
>
> It must also be computed per meter, not as a global constant — first dates range from 2024-12-09 to 2024-12-29, though all 29 meters end on 2026-07-11.

Reconciliation counts the implementation must reproduce

QUANTITY	TARGET	SOURCE
Validated interval rows	4,260,849	§8 · verified live
Total validated consumption	2,020,697.5 kWh	§8 · verified live
Null-consumption rows (29 first-valid + 28 ignored)	57	§11.2 · verified live

QUANTITY	TARGET	SOURCE
Calendar meter-day rows	16,704	§11.2
Meter-days with actual readings	14,753	verified live
Inserted missing meter-days	1,951	§11.2
Boundary meter-days	58	§11.2 · **verified live**
Long-gap-touched meter-days	2,421	§11.2
Meter-days passing the screen	14,228	§11.2 — the headline target

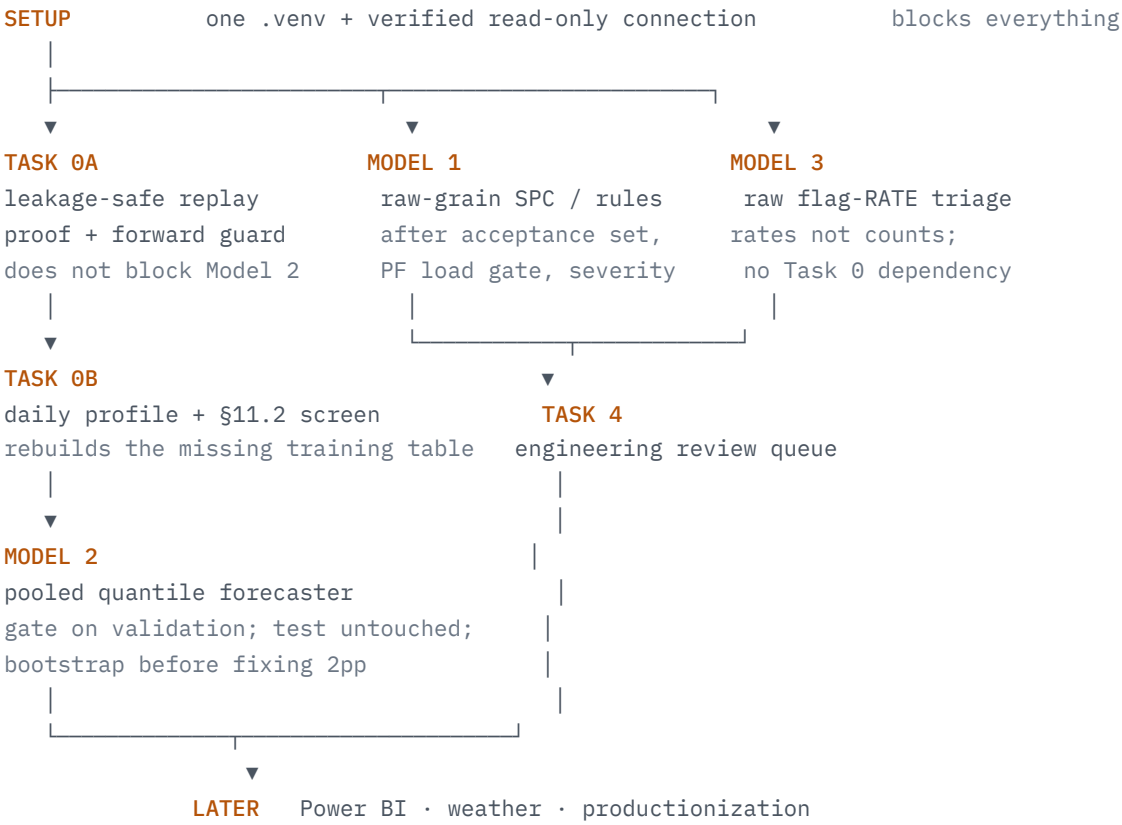
THE SCREEN IS A CONJUNCTION, NOT A SUBTRACTION

The three exclusion categories sum to 1,951 + 58 + 2,421 = **4,430**, but only 16,704 − 14,228 = **2,476** meter-days are actually excluded. The categories therefore overlap by **1,954** meter-days, so the reconstruction must evaluate all six conditions as a conjunction over each meter-day and never subtract category counts.

The largest overlap driver is ZN.17-PB's 201.36-day gap (2025-05-18 → 2025-12-05): those dates are simultaneously inserted missing days and long-gap-touched, because §11.2 marks *every* local date a gap spans, not just its endpoints.

Sequence

Final approved implementation order



Three tracks run in parallel after SETUP. The only hard serialization is 0A → 0B → Model 2, and 0A sits ahead of 0B for lineage rather than correctness — 0B consumes 0A's interval output so the daily table has a single provenance, cross-checked against the validated view.

Status — Addendum only. No models implemented, no Task 0 code written, nothing written to the database. Corrections 01 and 05 were verified against the live `sadinhub` database over a read-only connection; Bronze, Silver, Gold and Power BI are untouched.

Supersedes §03 (dependency graph), §04 (Task 0 row of the object table) and §09 (blockers) of the main architecture review. All other sections stand.